



Virinchi College
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Lab Report 1: Environmental Setup

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Imaging

Introduction

In this lab session, we will explore the process of creating a forensic image—a complete, bit-for-bit copy—of digital storage devices using FTK Imager. This is a vital step in digital forensics, as it helps maintain the original data's integrity and ensures the evidence is legally valid for courtroom use.

Description

In digital forensics, **imaging** means making a precise copy of data from storage devices like hard drives, solid-state drives, USB sticks, or even live memory (RAM), without altering the original data.

Key points include:

- FTK Imager enables the creation of forensic images from both physical storage devices and logical areas such as folders or partitions, as well as volatile memory.
- Unlike standard file copying, imaging captures **everything**, including hidden files, deleted data, unused space, and file system metadata.

Concept

Forensic imaging is the secure duplication of data from a source device into a format that ensures the information is unchanged and legally valid. FTK Imager accomplishes this through the following features:

1. Sector-by-Sector Duplication

- The tool reads each sector of the source storage medium to generate an exact and complete image, capturing all data—visible and hidden.

2. Forensic Integrity

- It calculates hash values (MD5, SHA-1, SHA-256) for both the original and the copy, allowing forensic examiners to confirm that the duplicate is identical to the source.

3. Image Formats Supported

- FTK Imager supports several widely accepted forensic image formats, including RAW/DD, E01 (used by EnCase), AFF, and AD1. It also allows users to compress images and split large files as needed.

4. Preview and Selective Imaging

- The tool provides a built-in viewer to look at files before capturing them and offers the ability to choose specific files or folders for imaging using the “Custom Content” option.

5. Chain-of-Custody Documentation

- Users can add case-related information (e.g., examiner name, case number) to the image file and generate logs that track imaging steps and hash values, ensuring the evidence can be trusted in legal settings.

Output

This section covers the configuration and use of FTK Imager, as well as the resulting forensic image that can be used for further investigation or presented in court as reliable evidence.

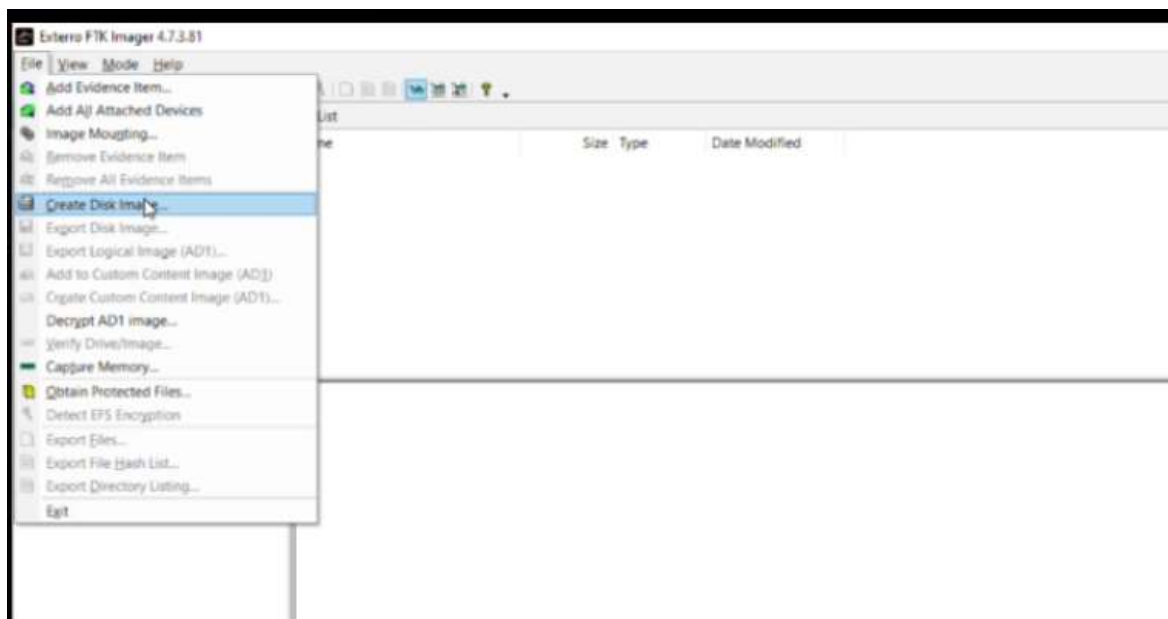


Image 1: Click on File and then “Create Disk Image”

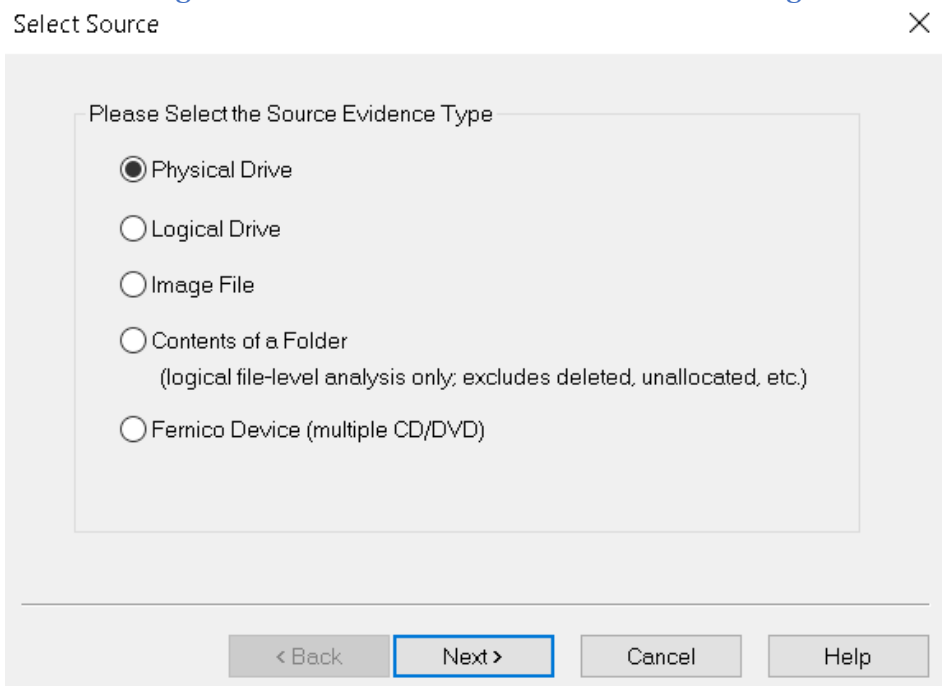


Image 2: Select Physical Drive

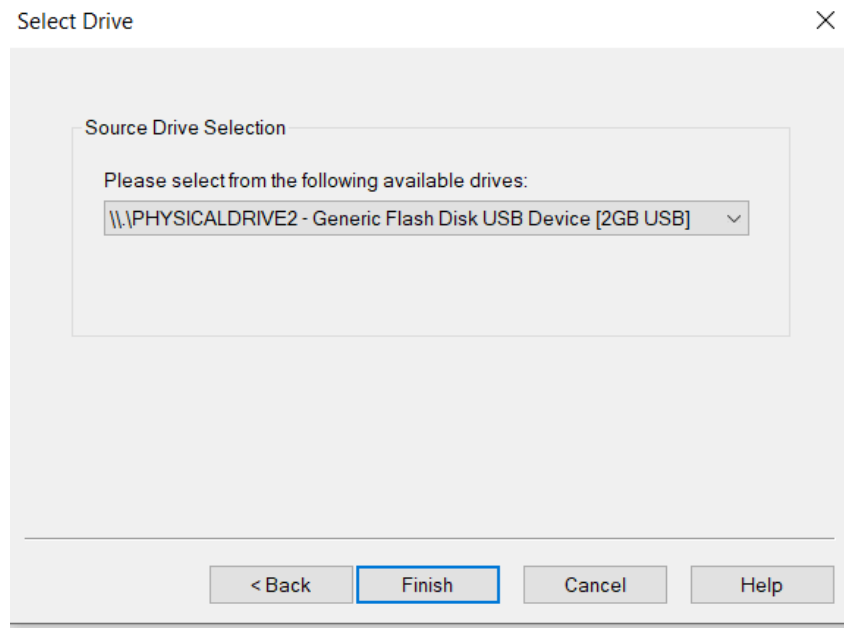


Image 3: Select the Imaging Disk

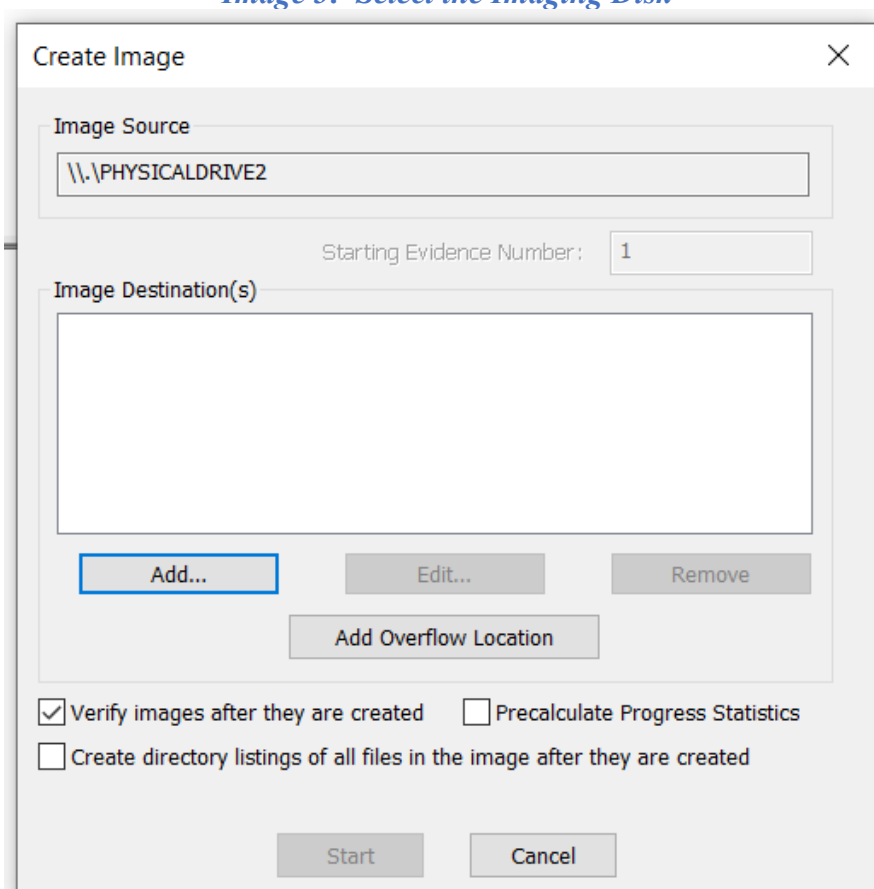


Image 4: Click on Raw

Select Image Type ✕

Please Select the Destination Image Type

☒ Raw (dd)

☐ SMART

☐ E01

☐ AFF

< Back Next > Cancel Help

Image 4: Select Raw and Press Next

Evidence Item Information ✕

Case Number :

Evidence Number :

Unique Description :

Examiner :

Notes :

< Back Next > Cancel Help

Image 5: Fill Up the Details

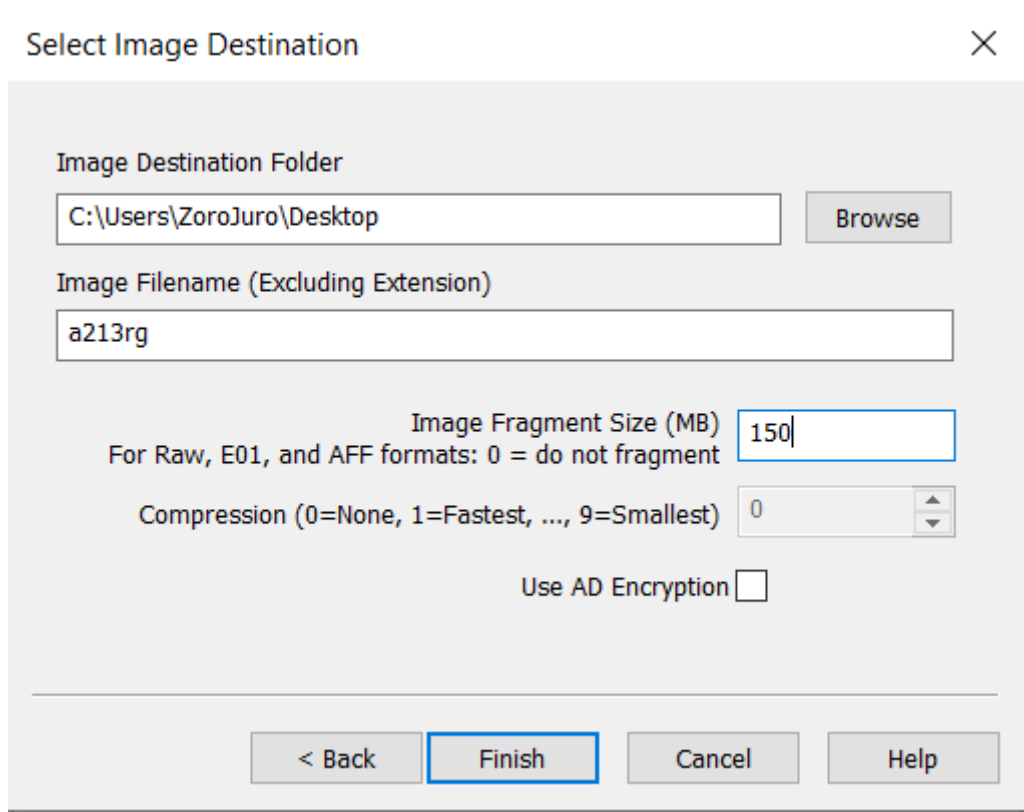


Image 6: Name and Destination Folder

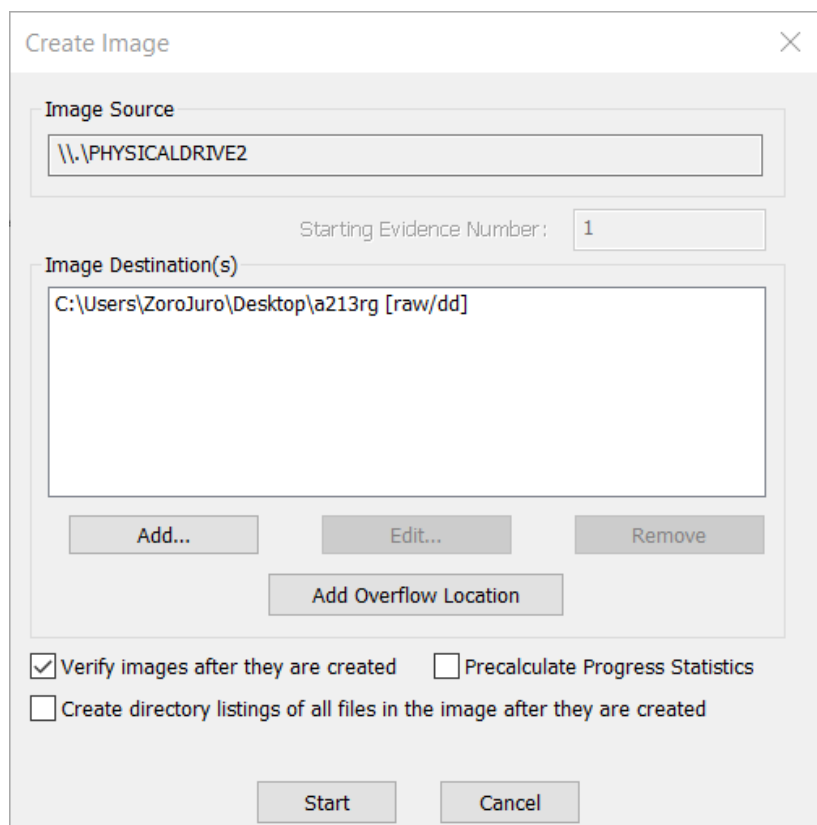


Image 7: Select and Start

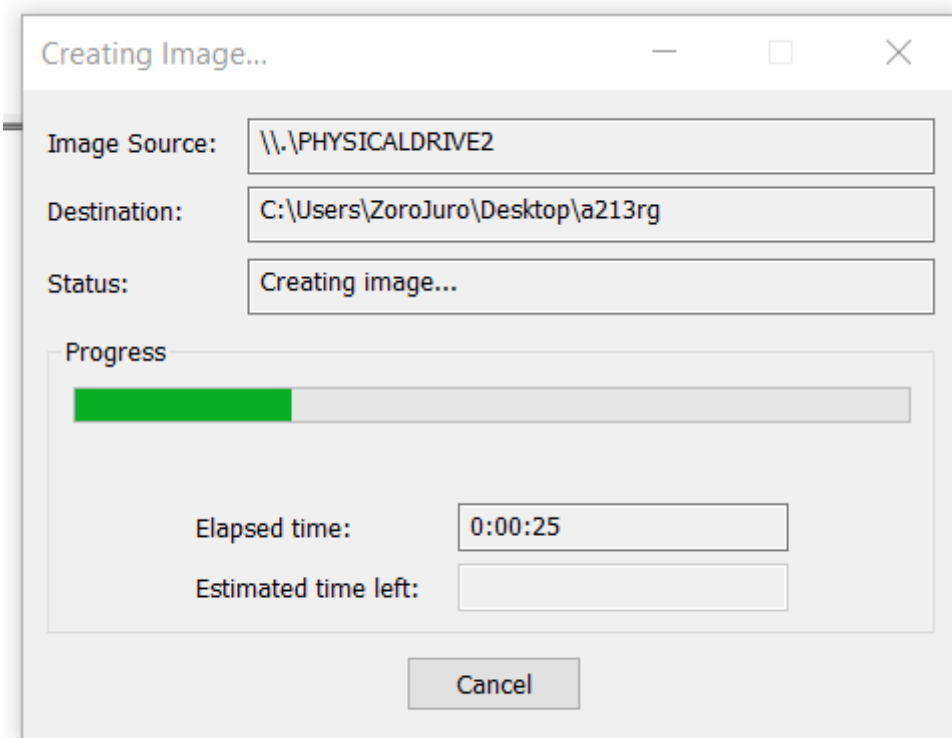
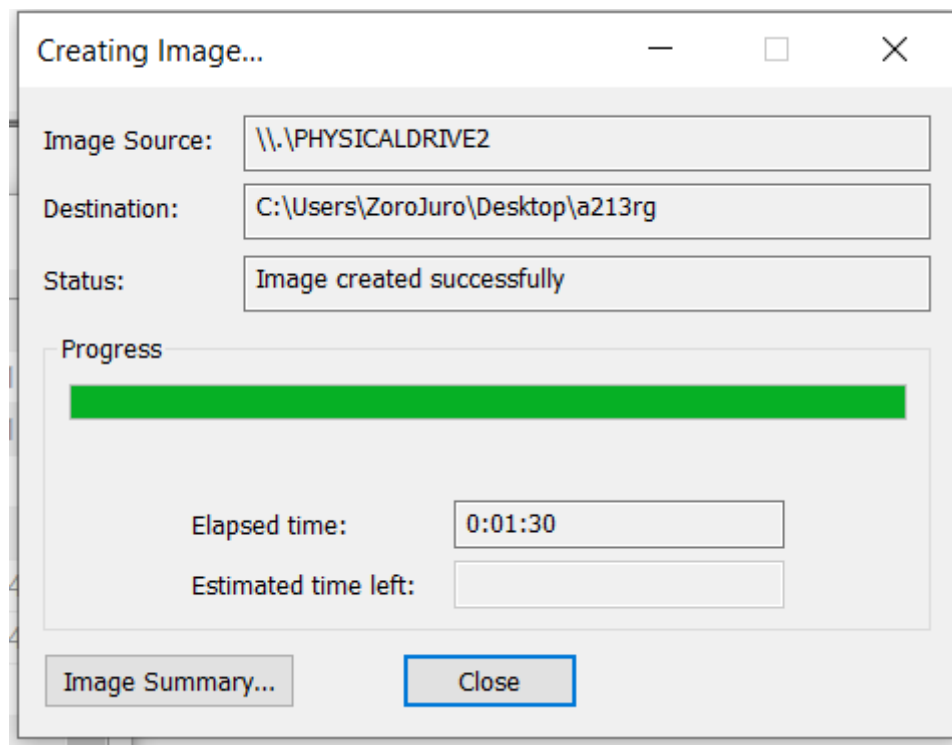


Image 8: Wait for Imaging to Finish



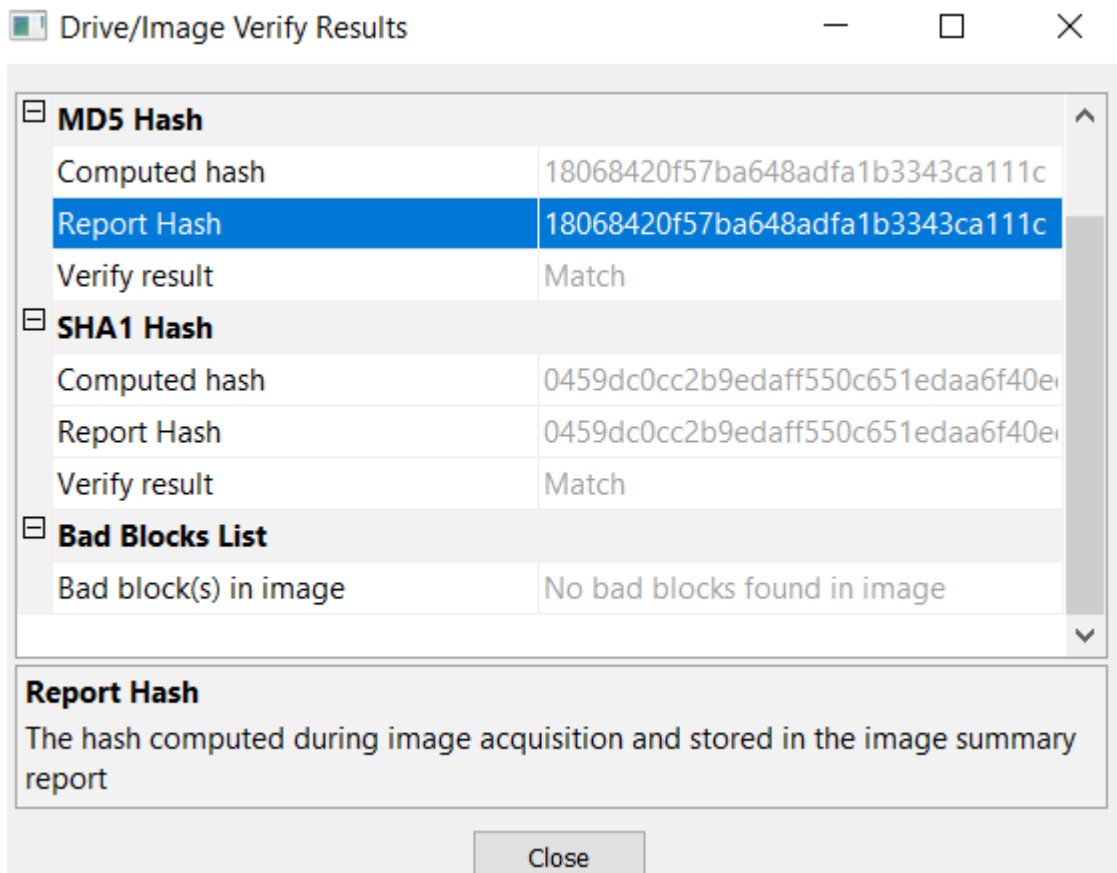


Image 9: Verifying and Completed

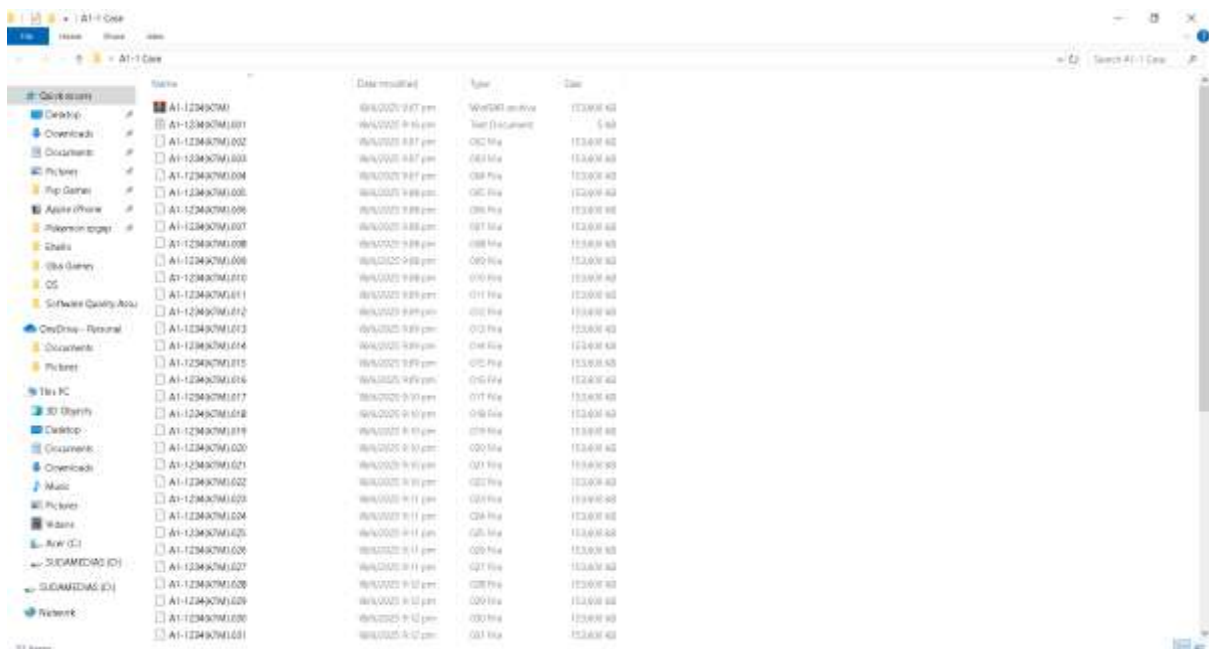


Image 10: Folder where Imaged Disk was Stored

Conclusion/Discussion

The imaging process plays a vital role in digital forensic investigations as it ensures the preservation of evidence without altering the original data. It captures information in a complete, tamper-proof, and verifiable way. FTK Imager stands out by offering both full and selective imaging options, along with strong hash-based validation and detailed documentation. However, while it's highly effective for data acquisition, it does not provide advanced analysis features. For in-depth investigation and artifact interpretation, tools like FTK, EnCase, Autopsy, or X-Ways are typically used.